034IN - FONDAMENTI DI AUTOMATICA - FUNDAMENTALS OF AUTOMATIC CONTROL ADDITIONAL PARTIAL WRITTEN EXAMINATION A.Y. 2023/2024

September 16, 2024

First and Last Name:	
Group:	Group A
Exercise:	Exercise 1 - 2 Questions
Available Time Frame:	60 minutes

Important Notes:

- Write your answers on a single white sheet of paper using a black pen.
- Do not write in blue ink or pencil.
- Do not hand in additional sheets.
- The clarity and precision of your answers will be subject to evaluation.

I declare that the answers to this exercise are my own and only my own work and that I have not consulted with others.

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Exercise: Exercise 1

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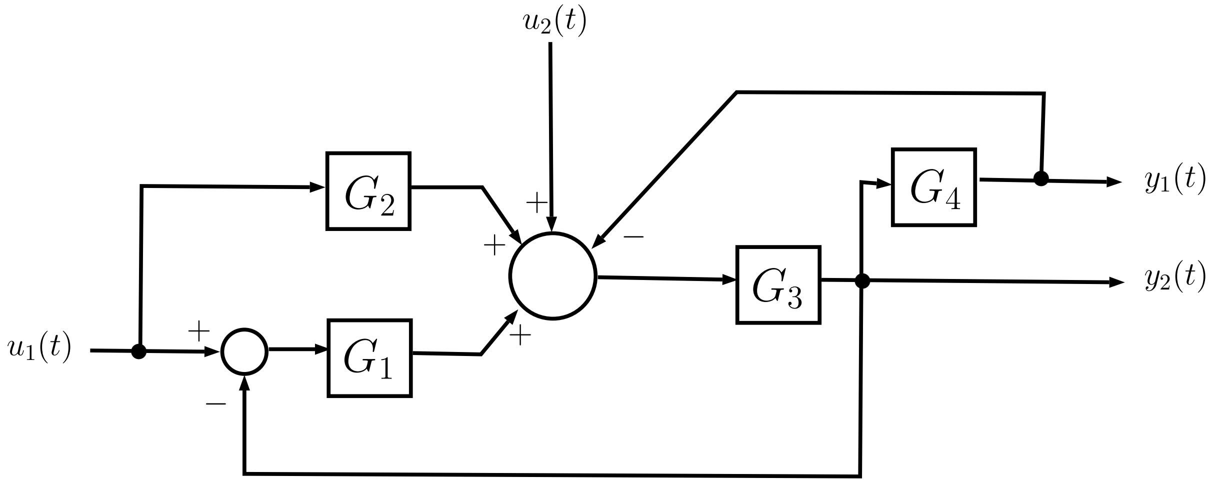
Exercise 1

The closed-loop system shown in the block diagram has two input signals, namely $u_1(t)$ and $u_2(t)$, and two outputs, $y_1(t)$ and $y_2(t)$.



Question 1.1.

- Find the transfer functions $T_{u_1}^{y_2}(s)$, from the input $u_1(t)$ to the output $y_2(t)$, and $T_{u_2}^{y_1}(s)$ from the input $u_2(t)$ to the output $y_1(t)$.

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Exercise: Exercise 2

Available Time Frame: 60 minutes

Exercise 2

Consider an LTI system with the following characteristic polynomial:

$$p(s) = s^3 + (5 + K)\ s^2 + (7 + 2\,K)\ s + (3 + 2\,K)$$

where $K \geq 0$.

Question 2.1.

- Analyse the system's stability as a function of the parameter K.
- What are the roots of the polynomial $p(s)$, when $K = 0$?
- Is there a value of K such that at least one of the roots of the polynomial is $\bar{s} = -2$?
- Is there a value of K such that at least one of the roots of the polynomial is $\bar{s} = -4$?

Comment on your findings.